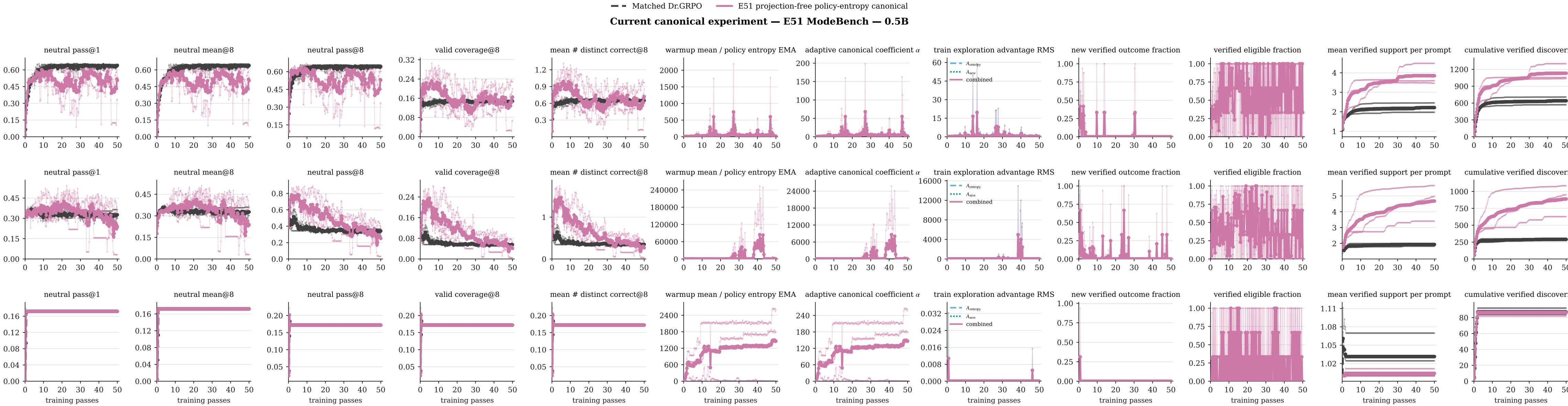


Python factors — E51 (live frontier)
graph coloring — E51 (live frontier)
Countdown — E51 (live frontier)



Each row uses its own matched Dr.GRPO and preregistered evaluation. Axes auto-zoom to each row's currently observed frontier and metric range; E51 uses K=8 exact-answer metrics with a 50-pass budget. Cancelled predecessor treatments are omitted.

Bank admission is validator-bound in every treatment: Countdown uses the executed exact AST, graph coloring the constraint-checked vector, and Python factors the externally executed return vector. The model cannot supply its own canonical key.

E51 freshly matches passive-tracking Dr.GRPO against projection-free policy-entropy adaptation: after 64 warmup observations, alpha is 0.10 times the run's warmup mean divided by the entropy EMA. Thin lines are seeds; heavy means appear only once all paired seeds 43/44/45 have reached an evaluation boundary.